

Yibin Zhen

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EDUCATION

University of Southern California

Expected Graduation: May 2029

Computer Engineering and Computer Science, B.S.

Relevant Coursework: Data Structures & Algorithms*, Distributed Systems*, Embedded Systems, Calculus (II, III), Linear Algebra, Differential Equations, Discrete Mathematics

* Currently enrolled

WORK EXPERIENCE

Pfizer (via Extern)

Remote

Applied Data Processing Extern

July 2026 – Present

- Built and benchmarked a **multi-engine OCR pipeline** (Tesseract, PaddleOCR, EasyOCR) to extract and layout-parse text from scanned pharmaceutical Supplier Data Files (SDFs), then designed a **rule-based document classification and routing system** that sorted 3+ document types into the correct extraction pipeline with **95%+ routing accuracy**.
- Engineered an end-to-end **Retrieval-Augmented Generation (RAG) pipeline** in **Python** using **LlamaIndex**, tuning chunking strategy and metadata filtering, then benchmarked open-source LLMs (**Mistral, Phi-2**) against proprietary models – **improving retrieval relevance by 30%+** – and shipped the system as an interactive **Gradio RAG chatbot**.

Mission Bit

Oakland, CA

Web Development Intern

Jun – Aug 2024

- Developed and shipped a full-stack career-exploration platform for **15+ high school students**, presenting the finished product at Verizon HQ.
- Engineered a **responsive, cross-device UI** achieving **less than 3 reported layout or rendering bugs** across desktop and mobile breakpoints.

PROJECTS

Robinhood Agentic Trading | Python, Claude API, Robinhood API, Tkinter

- Designed and deployed a fully autonomous trading system integrating Claude Sonnet's decision-making with Robinhood's live trading API, engineering real-time stop-loss monitoring and risk controls to manage live capital.
- Built a multi-signal pipeline synthesizing live price data, technical indicators, and RSS news from 10 national sources into LLM-driven trade decisions, run as a **1-month live pilot**.

Object Identifier | Python, OpenCV, Claude API

- Engineered a real-time object detection pipeline integrating Claude's vision API with OpenCV, achieving **80% detection accuracy at 30 FPS** on live webcam input with labeled bounding-box overlays.
- Implemented a throttled, configurable-interval API-call architecture, **cutting Claude API call volume and cost** without sacrificing real-time playback smoothness.

“untitled shooter game” (Roblox) | Lua

- Implemented a **modular shooting mechanic** from scratch – projectile simulation, hit detection, and server-side damage validation – **stable under real-world load of 100+ concurrent players**.
- Designed a **client-server system** using Roblox RemoteEvents with server-side anti-cheat validation, preventing client-side exploits.

TECHNICAL SKILLS

Technical Skills: C/C++, Python, HTML/CSS, Lua

Frameworks & Libraries: OpenCV, Tkinter

Tools & Technologies: Github, LLM APIs, Replit, Roblox Studio

Honors: QuestBridge National College Match Finalist, Oakland City Council Honors Commendation